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MOLECULAR BIOLOGY

Structures of *Vibrio cholerae* transcription complexes reveal how ToxR and TcpP recruit the RNA polymerase and activate virulence genes

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Activation of virulence in *Vibrio cholerae*, the etiological agent of cholera disease, is mediated by two transmembrane one-component signal-transduction proteins, ToxR and TcpP, which are also transcription factors. Using cryo–electron microscopy, we have solved five structures of the *ompU* and *toxT* transcription activation complexes, including the RNA polymerase (RNAP) holoenzyme, promoter DNAs, transcribed RNA, and their corresponding transcription factors, ToxR or TcpP and ToxR-TcpP, respectively. Activation is achieved through the interaction of ToxR or TcpP with the α–C-terminal repeat domain of RNAP where a single residue of the activator, a phenylalanine, appears to be the most critical contact, as confirmed by mutagenesis. No interactions of the transcription factors were observed with other subunits of the RNAP, i.e., the σ subunit as it occurs in the structurally related PhoB family of two-component transcription factors. The structures, and their comparison with our previously solved DNA promoter–ToxR x-ray structures, unveil the molecular mechanism of cholera virulence gene activation.

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INTRODUCTION

Cholera is an acute gastrointestinal diarrheal disease caused by the ingestion of the pathogenic bacterium *Vibrio cholerae* from contaminated water or food. Between 1.3 and 4.0 million cases of cholera, and 21,000 and 143,000 deaths, have been estimated yearly worldwide (*1*). The World Health Organization has risen concerns about an escalation in global cases and deaths associated to cholera in recent years (*2*). This global resurgence has been classified as a grade 3 emergency in January 2023, the highest level of emergencies.

Once the bacterium reaches the small intestine, the presence of bile salts and anaerobic conditions trigger a regulatory cascade in *V. cholerae* that results in the expression of the cholera toxin (CT) and the toxin coregulated pilus (TCP) (*3*, *4*). The expression of *ctx* and *tcp* operons is activated by ToxT, which is positively regulated by ToxR and TcpP, two transmembrane one-component signal-transduction proteins, acting as transcription factors by their cytosolic N-terminal DNA binding domains (DBDs). Both transcription factors bind to the *toxT* promoter (*5*). Furthermore, ToxR alone is also involved in *V. cholerae* resistance to bile salts, because it down-regulates the expression of the membrane porin OmpT and up-regulates the expression of OmpU, which blocks bile salts intake (*6*, *7*).

The *V. cholerae* virulence activation cascade has been thoroughly studied to determine where and how ToxR and TcpP transcription factors bind to the *ompU* and *toxT* promoters (*6*, *8*–*15*). However, little is known about how the RNA polymerase (RNAP) is recruited, the differences among the activation mode in each promoter, or the interactions between the RNAP and these transcription factors. Using single-particle cryo–electron microscopy (cryo-EM), we have solved five structures of the *ompU* and *toxT* transcription activation complexes (TACs), including the RNAP holoenzyme, the DNA, and their corresponding transcription factors, ToxR or TcpP and ToxR-TcpP, respectively. All these structures, together with our previous work on the immediately upstream region of both promoters (*12*), allow us to propose a model of the virulence-related transcriptional cascade activation in *V. cholerae*, after the bacteria reach the intestine.

V. cholerae transcription activation structures in this study present how two different transcription factors, ToxR and TcpP, whose activation mechanisms and their structures in complex remained elusive, activate in a similar fashion the virulent gene cascade upon host infection. The architecture of the complexes involves the *V. cholerae* RNAP holoenzyme structure (2α, β, β′, ω, and σ^D), a DNA promoter fragment with the transcription factors bound and in contact with the RNAP. Both transcription factors display similar, but not identical, activation mechanisms, where slight differences in the structures, interaction, and fine-tuning mechanisms result in different promoter selection for the *ompU* promoter, and a collaborative activation for the *toxT* promoter.

The aim of this work was to unveil how transcription is activated by ToxR and TcpP transcription factors. We knew in advance that the structure of the components of the holoenzyme would be very similar to those of the *Escherichia coli* enzyme, since the sequence is highly conserved between these species of bacteria. What we did not know is whether the binding of the transcription factor would cause a rearrangement in some parts of the polymerase as occurs, for example, with PhoB (*16*). It does not, despite the similarity of ToxR

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and TcpP DBDs to PhoB DBD, and, thus, these transcription factors should be classified as different classes.

RESULTS

The first *V. cholerae* complexes that we analyzed by cryo-EM were the *ompU* open promoter complex (RPo-*ompU*) and the *ompU* transcription initiation complex (TIC-*ompU*) (Fig. 1A). Both complexes comprise the *V. cholerae* RNAP holoenzyme ($E\sigma^D$) bound to a 69–base pair (bp) DNA fragment of the *ompU* promoter, which includes the ToxRBox operator (the DNA region recognized by ToxR), and a 13-bp length transcription bubble. The formation of the TIC (Fig. 1A; Materials and Methods) was achieved by the addition of an extra five-nucleotide RNA fragment (5-mer RNA). The processed datasets rendered maps at 3.0-Å resolution for the RPo and 2.6-Å resolution for the TIC (figs. S1 and S2; Materials and Methods). The *V. cholerae* RNAP structure presents the classical and highly conserved bacterial architecture involving 2α, β, β′, and ω subunits disposition at the core, and the addition of a dissociable σ factor (σ^D, equivalent to σ^{70} in *E. coli*) (Fig. 1A). It shows a high amino acid sequence identity (ranging from 77 to 90%, depending on the subunit) and structural similarity [overall root mean square deviation (RMSD) of 0.739 Å] with its *E. coli* ortholog (*17*). The interaction of the *V. cholerae* RNAP with the DNA in the RPo complex is similar to the one described for the *E. coli* RNAP (*18*, *19*), where the σ^D factor interacts with the putative −10 and −35 elements while the α–C-terminal repeat domain (CTD) interacts with the upstream promoter element (UP). The higher resolution in the area around the active site in the TIC allowed us to use it as a scaffold to build all the subsequent complexes, which include the ToxR and TcpP transcription factors, as well as to allocate the nascent 5-mer RNA annealed with the DNA template.

The formation of the TAC (3.1-Å resolution; fig. S3) on the *ompU* promoter was achieved by the addition of the cytoplasmic domain of the transcription factor ToxR ($ToxR^{cyt}$) together with the RNAP to the 69-bp transcription bubble (TAC-*ompU*) (Fig. 1A). For the *toxT* promoter, we solved two structures of the activation complex, differing in the promoter fragment length. The TAC-*toxT*-S ("S" for short) (2.7-Å resolution; fig. S4) was reconstructed with a 71-bp *toxT* promoter fragment, including a 13-bp transcription bubble mismatch, and the two TcpP binding sites (TGTAA), plus the cytoplasmic domain of the TcpP transcription factor ($TcpP^{cyt}$) (Fig. 1B). The TAC-*toxT*-L ("L" for long) (2.9-Å resolution; fig. S5) assembly has a longer promoter DNA fragment (110 bp), including the TcpP and ToxR binding sites, as well as both transcription factors (Fig. 1B), as explained in Materials and Methods.

The *ompU* TAC involves two ToxR molecules, the α-CTD and $\sigma^D{}_4$ domains of the RNAP holoenzyme

The TAC-*ompU* shows the key ToxR-DNA interactions for promoter activation (Fig. 2). Two ToxR protomers bind in tandem, with the recognition helix fitted into the DNA major groove, and the wing interacting with the minor groove (Fig. 2A). ToxR targets AT-tract sequences, with a narrow minor groove (*12*), and recognizes the bases thymine and adenine with residues Threonine 77 (T77) and Q78 of the recognition helix, respectively (fig. S6). Previous studies have demonstrated the critical nature of ToxR T77 and Q78, since alanine mutations at these positions disrupt *ompU* activation. Furthermore, a T77A mutation also completely disrupts *toxT* activation, and a previously found Q78R mutation affects promoter binding. In addition, ToxR R84 is also critical for binding, since this residue interacts via a salt bridge with a DNA phosphate (*12*, *14*).

ToxR interacts with the RNAP through an activation interface that involves the α-CTD, which contacts the σ^D domain 4 ($\sigma^D{}_4$) of the RNAP holoenzyme (Fig. 2B). ToxR #2 molecule binds upstream, covering the promoter positions from −49 to −42, followed by ToxR #1 interacting from bases −40 to −35 (Fig. 2C). In addition, ToxR #1 has a contact surface with the RNAP α-CTD, the latter covering the promoter region from −44 to −38. Thus, the α-CTD is the nexus between the transcription factor and the body of the RNAP, since it interacts simultaneously with the UP element, with one ToxR and with the $\sigma^D{}_4$ positioned in the DNA in the opposite face of the ToxR molecules (Fig. 2C). Our final cryo-EM atomic model lacks the second α-CTD, since its density was only visible at low map threshold and in some of the best two-dimensional (2D) classes, where it appears to interact with molecule ToxR #2 in a similar way as the first α-CTD does with ToxR #1 (fig. S7).

When comparing the ToxR-*ompU* promoter crystal structure [Protein Data Bank (PDB): 8B4B] (*12*) and the cryo-EM structure presented in this work, the latter presents two $ToxR^{cyt}$ molecules bound in tandem, while there are four ToxR protomers in the crystal structure, two of them coincident with the ones in the RNAP complex and another tandem in an inverted repeat mode with respect to the first pair (Fig. 2C). This second pair would not be able to bind the DNA in the RNAP complex because one of the molecules would clash with the RNAP $\sigma^D{}_4$ and α-CTD. The position of the ToxR shared in both structures is identical, even though a few contacts are different between ToxR #1 and the DNA, indicating that the readout mechanism lies mostly on the specific DNA topology (Fig. 2E). ToxR-ToxR contacts are also conserved in both structural approaches. ToxR #2 wing is docked into the DNA minor groove while it interacts with ToxR #1, acting as a stabilization staple and increasing the interaction surface (Fig. 2, A and B). The differences in the binding between ToxR #2 and ToxR #1 arise from the DNA structure through the kink at position −34 between two rather straight stretches (Fig. 2E). The DNA, bent by the RNAP, presents a minor groove widening, followed by a straight stretch in the DNA duplex central axis (fig. S8A), which does not allow ToxR #1 wing domain to interact with this region. Therefore, ToxR #2 stabilizes ToxR #1 through wing contacts and facilitates the ToxR #1–α-CTD interaction. These interactions are consistent with a model for *ompU* transcription activation where ToxR molecules already bound to the DNA, away from the transcription start site, facilitate additional ToxR binding events, progressing downstream toward the transcription start site (*15*). The *ompU* promoter wing–based ToxR-ToxR interactions therefore promote the transcription factor recruitment that lastly leads to the binding of one ToxR molecule contacting the RNAP α-CTD for the transcription activation.

ToxR class I activation mechanism is based on an α-CTD rotation leading to enhanced binding

The transcription activation interaction occurs between the ToxR transactivation loop and the RNAP α-CTD, and it involves a total of twelve residues, where five belong to the transcription factor and seven to the RNAP α-CTD, according to a PISA (Proteins, Interfaces, Structures and Assemblies) analysis (Fig. 3, A and B) (*20*). Alanine substitutions in the ToxR transactivation loop were described previously (*14*), where three out of five residues of the loop (G68A, F69A,

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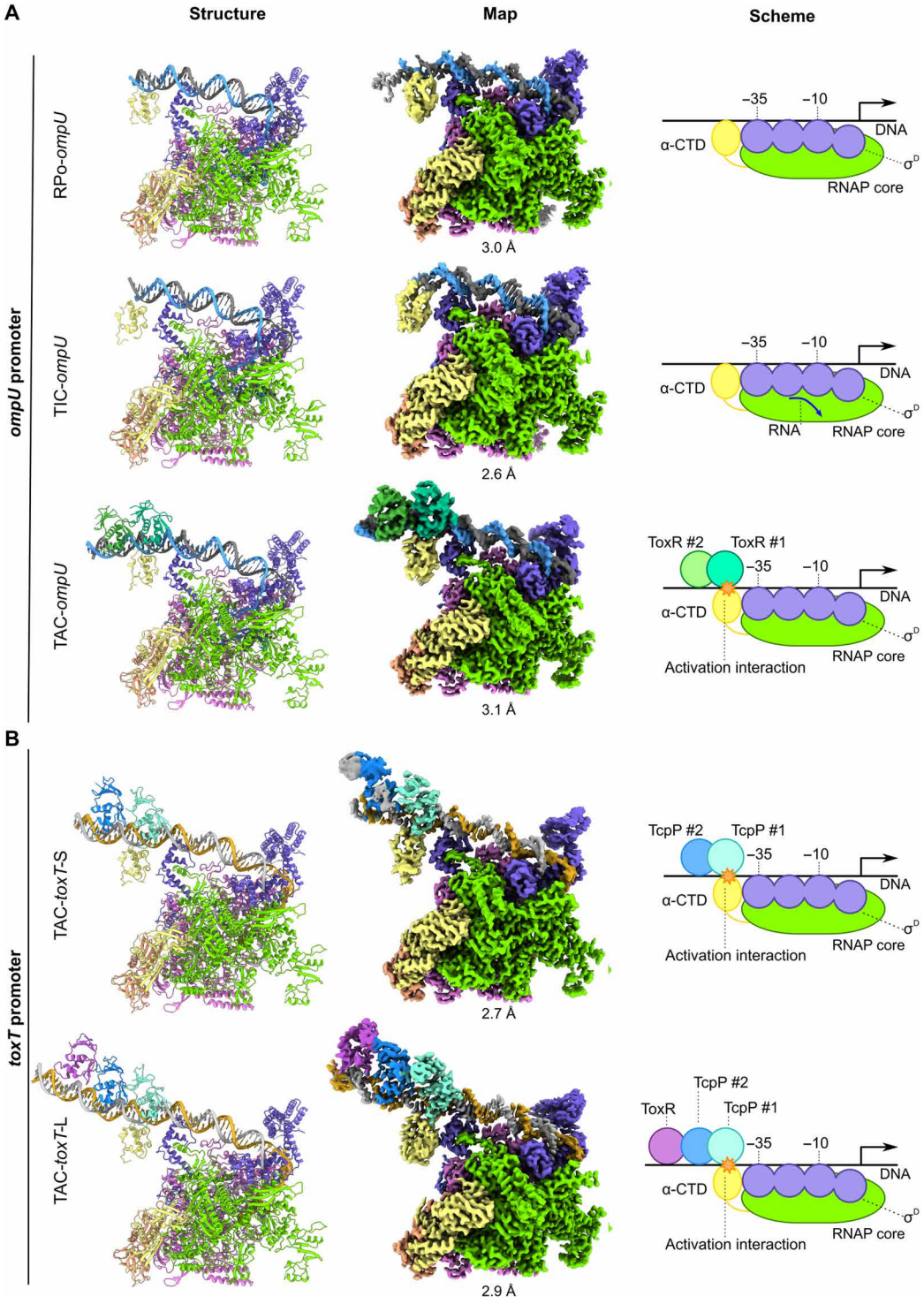

Fig. 1. Overview of the *V. cholerae ompU* and *toxT* transcription complex cryo-EM structures. (**A**) RPo-*ompU* (top), TIC-*ompU* (middle), and TAC-*ompU* (bottom) structures represented in ribbon (left), cryo-EM maps (center), and schematic representations (right). The RNAP subunits are colored as follows: α1, pale yellow; α2, orange; β, lawn green; β′, pink; ω, medium purple; σ^D, indigo; ToxR #1, spring green; ToxR #2, lime green; nontemplate DNA, gray; template DNA, corn blue; RNA, dark slate blue. The maps were colored in ChimeraX (*42*), using the color map function according to a structure with a 10 Å distance radius. ToxR #1 and ToxR #2 bind to the *ompU* DNA promoter at the ToxRBox and interact with the α-CTD in TAC-*ompU*. (**B**) Short (TAC-*toxT*-S) (top) and long (TAC-*toxT*-L) (bottom) *toxT* TAC structures represented in ribbon (left), cryo-EM maps (center), and schematic representations (right). The RNAP subunits α, β, β′, ω, and σ^D are colored like in (A), whereas the nontemplate DNA is goldenrod; template DNA, silver; TcpP #1, aquamarine; TcpP #2, dodger blue; ToxR, medium orchid. Two TcpP protomers are bound at their canonic DNA binding sites in TAC-*toxT*-S, interacting with the α-CTD. On TAC-*toxT*-L, the structure shows two TcpP protomers bound to the DNA, with an additional ToxR molecule upstream TcpP.

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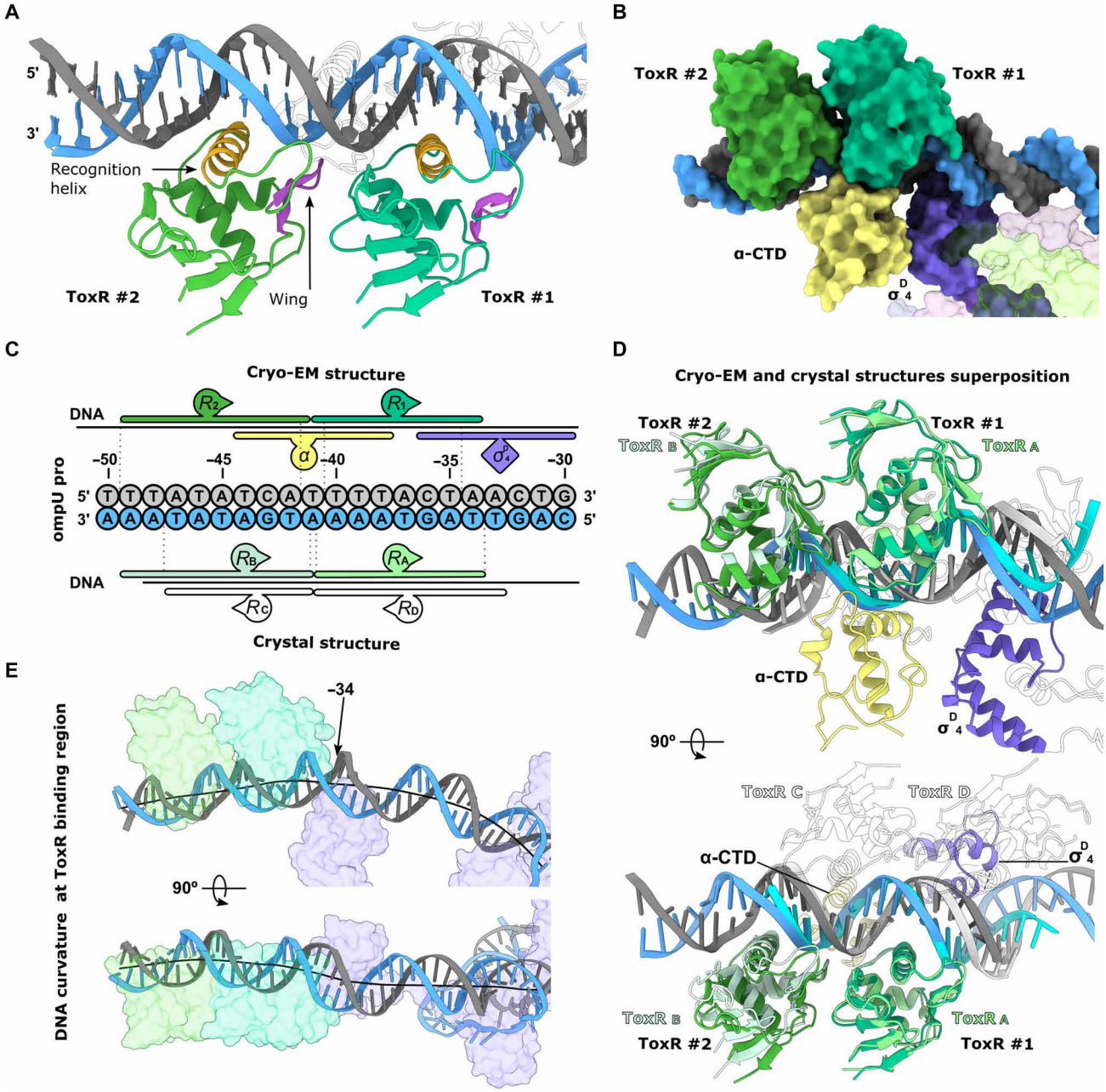

Fig. 2. The *ompU* TAC involves two ToxR molecules, the α-CTD and the σ^D domains of the RNAP holoenzyme. (**A**) Ribbon representation of the ToxRBox operator in the TAC-*ompU* complex, with two ToxR protomers bound. The different subunits are colored as follows: ToxR #1, spring green; ToxR #2, lime green; nontemplate DNA, gray; template DNA, corn blue. All the other $E\sigma^D$ subunits are represented as silhouettes. The ToxR wing domain is highlighted in violet and the recognition helix in yellow. (**B**) Surface representation of the TAC-*ompU* TAC, where ToxR #1, ToxR #2, nontemplate DNA, and template DNA are color-coded as above. The α-CTD is depicted in pale yellow and the σ^D in indigo. (**C**) Schematic comparison of the DNA binding regions of the TAC-*ompU* from the cryo-EM structure (top) and the crystal structure (PDB: 8B4B) (*12*), with four ToxR protomers ($ToxR_{A,B,C,D}$). The straight line represents the DNA built in the structures. The DNA sequence in the middle of the figure is color-coded as in (A). The DNA positions are labeled relative to the transcription start site, and the dashed lines represent the DNA binding delimitation for ToxR #1 and #2 of the Cryo-EM structure and $ToxR_A$ and $ToxR_B$ from the crystal structure. (**D**) Superposition of the TAC-*ompU* structure and the crystal structure (PDB: 8B4B) (*12*) represented in ribbon. Subunits are color-coded as in (C). (**E**) TAC-*ompU* DNA bending. Ribbon representation of the template (corn blue) and nontemplate (gray) DNA strands. ToxR #1, ToxR #2, and the σ^D (transparent surface) are depicted to match the color code in (A) (other subunits are omitted for clarity). The DNA central axis is represented by a black line, with a kink, with a minor groove widening, at the −34 position.

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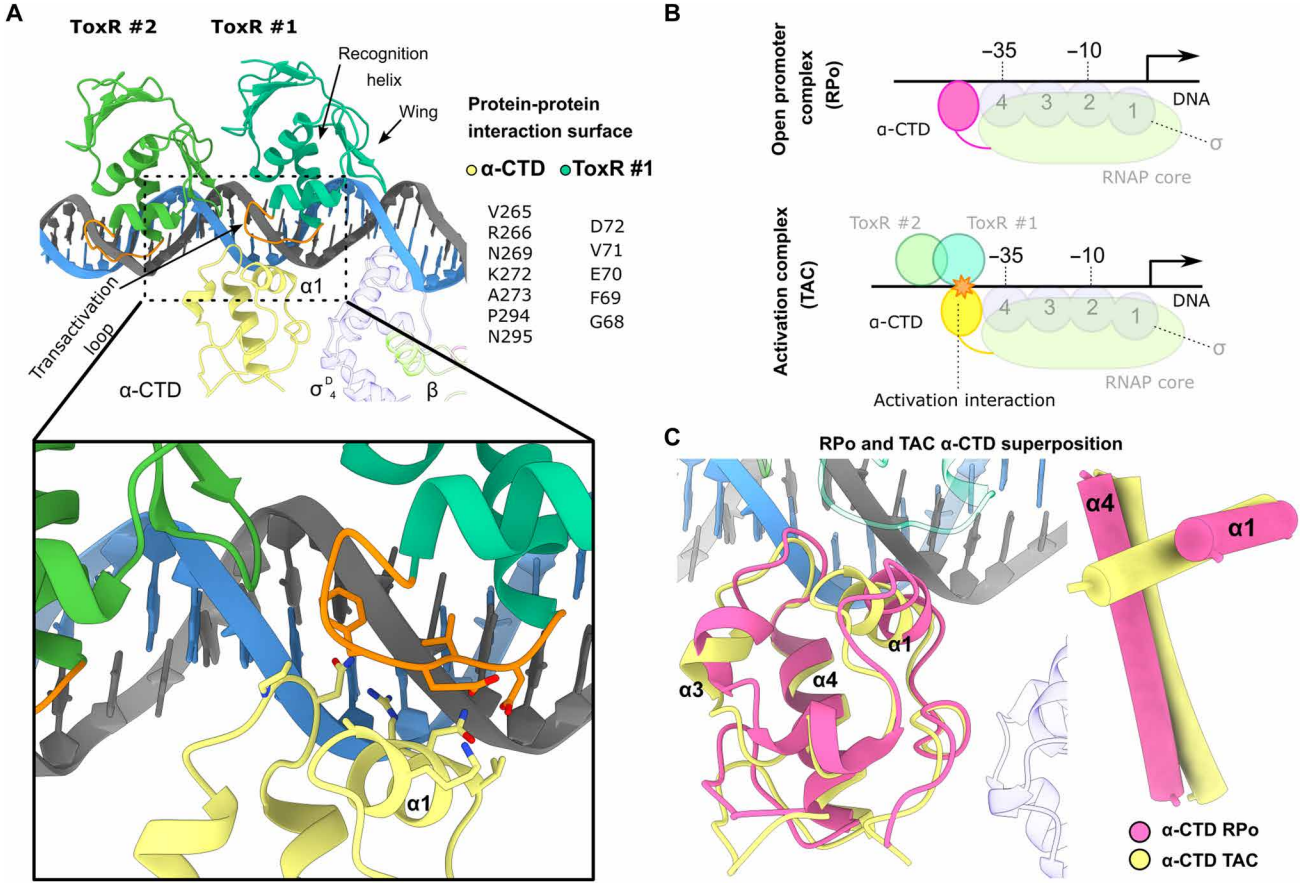

Fig. 3. ToxR class I activation mechanism. (**A**) Ribbon representation of the *ompU* ToxRBox DNA operator region in TAC-*ompU* structure. The subunits are colored as follows: ToxR #1, spring green; ToxR #2, lime green; α-CTD, pale yellow; β, lawn green; σ^D, indigo; nontemplate DNA, gray; template DNA, corn blue. The ToxR #1–α-CTD interaction residues are listed on the right and represented as sticks at the zoom panel below. (**B**) Schematic representation of the RPo-*ompU* and TAC-*ompU* complexes (**C**) Structural superposition of the RPo-*ompU* and TAC-*ompU* complexes, focused on the α-CTD. The RPo-*ompU* α-CTD is represented in pink and the TAC-*ompU* α-CTD in pale yellow. A tube representation of the α1 and α4 helices displacement when ToxR is present is shown on the bottom right panel.

and V71A) that we find to be in contact with the RNAP α-CTD were shown to disrupt *ompU* activation.

A superposition of the RPo-*ompU* and TAC-*ompU* structures shows a small rotation of the α-CTD upon $ToxR^{cyt}$ binding, causing a displacement of helix α1 and loop α3-α4 (Fig. 3C). The α-CTD binds to the DNA with helix α1 positioned over the DNA minor groove (*21*). The α-CTD–DNA interface area is 447.4 $Å^2$ in the RPo-*ompU* complex. In contrast, the α-CTD in the TAC-*ompU*, also fastened by its α-CTD–ToxR contacts, has an overall anchoring surface of 617.4 $Å^2$. Thus, there is a 37.9% increment of the α-CTD interaction surface in the activation complex. Furthermore, the interaction surface between the α-CTD and the $\sigma^D{}_4$ at the TAC-*ompU* is 150.7 $Å^2$, 17.3% larger than in the RPo-*ompU* with 128.4 $Å^2$, showing again a better attachment of the α-CTD in the activation complex.

The cryo-EM maps of both RPo-*ompU* and TAC-*ompU* datasets provide further evidence of the α-CTD stabilization when ToxR is present. On the RPo-*ompU* map, the α-CTD density disappeared during the first refinement step. A further calculation, using a more homogeneous set of particles, allowed the visualization of the domain (fig. S1). Nevertheless, the presence of the α-CTD density is clear after the first refinement in the TAC-*ompU* volume (fig. S3), suggesting that the α-CTD binding to the DNA promoter UP element is more stable.

TcpP uses a winged helix-turn-helix motif to bind DNA

The TAC-*toxT*-S complex (Fig. 4A) displays two $TcpP^{cyt}$ molecules bound in tandem to the −52 to −35 region of a 71-bp *toxT* promoter fragment. The cryo-EM maps show well defined density for the downstream $TcpP^{cyt}$ protomer α helices and loops, but an imprecise view of the β-sheet cap on top. In the immediately upstream positioned $TcpP^{cyt}$, the density is well defined only in the regions with DNA interaction, that is, the recognition helix and the wing.

The TAC-*toxT*-L complex presents a third protomer, bound to the ToxRBox operator, upstream of the TcpP molecules (Fig. 4A). It was assigned to a $ToxR^{cyt}$ molecule, according to its nucleotide recognition region, determined by footprinting experiments, where no TcpP molecules should be bound (*11*). The TAC-*toxT*-L focused maps present better quality and allow a more appropriate characterization of the $TcpP^{cyt}$ and *toxT* promoter (fig. S9). Thus, the TAC-*toxT*-L is our structure of choice for the analysis.

$TcpP^{cyt}$ presents a winged helix-turn-helix (wHTH) fold with three α helices and a two–β strand wing, all covered with a β-sheet

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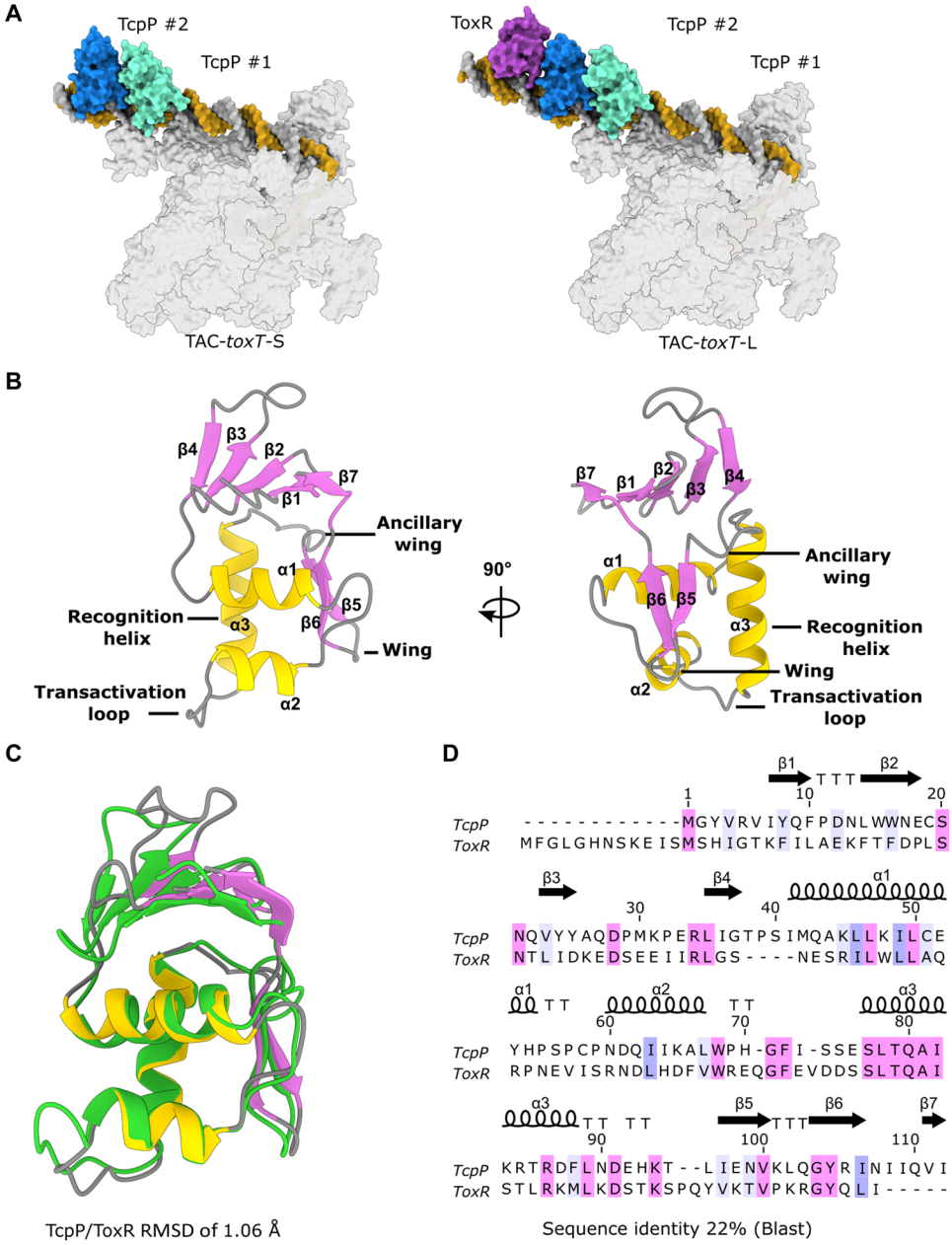

Fig. 4. TcpP uses the expected wHTH DNA binding motif. (**A**) Surface representation of the TAC-*toxT*-S (left) and TAC-*toxT*-L (right) complexes. The RNAP core is colored in transparent gray, TcpP^{cyt} #1 in cyan, TcpP^{cyt} #2 in dodger blue, and ToxR^{cyt} medium orchid. (**B**) Ribbon representation of two TcpP^{cyt} structure views rotated by 180°. The α helices are colored in yellow, whereas the β strands are in violet. The structure presents β1 to β4 strands followed by the helix α1 and an HTH motif formed by helices α2 and α3 (recognition helix) connected by the transactivation loop. Helix α3 is followed by the ancillary wing coiled coil and the wing, formed by two antiparallel β strands (β5 and β6). The C terminus ends with the β7 strand, which forms a β-sheet cap with β1 to β4 strands. (**C**) Structure superposition of the TcpP^{cyt} #2 [colored like in (B)] from the TAC-*toxT*-L complex against the ToxR^{cyt} (colored in green) (PDB: 8B4B) with an RMSD of 1.06 Å. (**D**) Protein sequence alignment (*51*) of TcpP^{cyt}, with the secondary structure elements depicted above the sequence, against ToxR^{cyt} (22% identity). The colored squares on the sequence represent the identity. Purple gradient is for chemical similarity between residues (light purple shows less identity and dark for higher) and pink for full identity.

cap (Fig. 4B). Helix α1 (residues 41 to 54) connects with helix α2 (residues 60 to 67), followed by the transactivation loop that connects with the recognition helix α3 (residues 77 to 88), fitted into the DNA major groove. This helix is followed by an ancillary wing (residues 91 to 96), which connects the recognition helix and the wing domain β strands (residues 97 to 107). The C terminus ends with a β strand that is part of the β-sheet cap, which is shown as a uniform low-resolution layer in the unsharpened density maps and hence could not be built. The superposition of the TcpP^{cyt} structure to the ToxR^{cyt} PDB: 8B4B (*12*) (RMSD of 1.06 Å) displays a similar fold and position of the key structural elements of the wHTH transcription factor family (Fig. 4C), even though the sequences corresponding to

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the cytoplasmic domain of both transcription factors share a low 22% sequence identity (Fig. 4D). However, the conservation in the recognition helix (α3) is higher than in the overall alignment, sharing the base-specific recognition residues threonine and glutamine. The wing domain sequences (β5 to β6) differ on the wing tip, the loop between the two beta strands of the wing (Fig. 4D).

TcpP also activates transcription by interacting with the α-CTD of the RNAP

The *toxT* promoter activation interface shown in the TAC-*toxT*-L complex involves, in the 5′ to 3′ direction, a ToxR^{cyt} protomer (ToxR) followed by a TcpP^{cyt} molecule (TcpP #2), which interacts with another TcpP^{cyt} (TcpP #1) (Fig. 5). The number assigned to the TcpP molecules indicate their proximity to the transcription start site. The residues involved in both interactions were analyzed with PISA (*20*) and are listed in fig. S8C. The calculated surface interface between ToxR and TcpP #2 is 285 Å^2 and between TcpP #2 and TcpP #1 is 278.6 Å^2.

TcpP #2 binds the DNA covering the region from −52 to −43 of the *toxT* promoter (fig. S10). The main protein-DNA interactions involve the recognition helix and the wing (Fig. 5). The recognition helix recognizes the −49 dA:dT base pair through residues E76 and Q80. Moreover, K83 interacts with the dT −47 and dT −49. Furthermore, T79 interacts with dA −48, whereas R84 does with dG −52 (Fig. 5). Bases dA −49 and dG −52 have been shown previously to be essential for *toxT* activation (*9*). Last, the R86 interaction with dT −47 phosphate, which anchors the helix in the DNA major groove, has also been described to be critical since an R86A mutant was completely defective for *toxT* activation and DNA binding (*11*). TcpP #2 establishes one more base-specific interaction at the wing tip with K101 recognizing dC −43 while it establishes hydrophobic contacts with the DNA backbone (Fig. 5). This is a notable difference between ToxR-DNA and TcpP-DNA interaction with the minor groove, since ToxR wing residue (P101) does not recognize any specific base, limited to the sandwich-like hydrophobic interaction of the Pro ring in the minor groove (fig. S6B).

TcpP #1 shows fewer interactions with the DNA than TcpP #2 due to the conformation of the DNA in both the TAC-*toxT*-S and TAC-*toxT*-L structures, which shows a kink at the −49 and −32 positions, bending the overall DNA shape (fig. S8B). The DNA conformation does not show any substantial change in the stretch proximal to the RNAP upon ToxR binding (TAC-*toxT*-L structure). The differences in the DNA between both structures S and L are only in the upstream DNA stretch involving the ToxR binding. It is impossible to determine whether these differences are due to the longer DNA sequence or to ToxR binding, since we do not have a long DNA sequence structure without ToxR for comparison. In any event, the change does not translate to the area were α-CTD interacts. Therefore, we have ruled out that a change in the DNA upon ToxR binding is responsible for activation.

TcpP #1 covers the region from −41 to −35, where its recognition helix binds the −39 and −38 bases with residues E76, Q80, and K83 (Fig. 5). Only the dA −38 has been shown to be essential for *toxT* activation (*9*). The shared interactions between TcpP^{cyt} protomers highlight the relevance of having a dA:dT base pair (with the dA being in the nontemplate DNA strand) in the TcpP binding site where the recognition helix fits in the major groove. Moreover, TcpP #1 has a slightly rotated orientation relative to the DNA compared to TcpP #2, since the wing is completely out of the minor groove, and, therefore, it does not recognize any base (Fig. 5). Thus, besides the recognition helix, TcpP #1 is stabilized by protein-protein interactions with TcpP #2, and protein-DNA interactions involving other elements, such as the ancillary wing and the recognition helix R86 residue (Fig. 5). Residues 70 to 74 of the TcpP #1 transactivation loop interact through hydrophobic contacts with the α-CTD helix α1 (residues 265, 269, 272, and 273) (Fig. 5), covering a surface area of 145.5 Å^2, in the same fashion that we have described for ToxR on the *ompU* promoter. Last, TcpP #1 and TcpP #2 differ in the transactivation loop interactions with the DNA (Fig. 5). While TcpP #1 interaction involves S74 and W68, TcpP #2 involves S74 and F72. These differences rely on TcpP #1 transactivation loop rearrangement for α-CTD interaction.

The ToxR^{cyt} domain can bind the TcpP DNA operator of the *toxT* promoter in the opposite orientation to the crystal structure

The TAC-*toxT*-L activation complex includes three transcription factors in a row. This arrangement differs from the one shown in the published crystal structure with two ToxR molecules bound to the *toxT* DNA fragment encompassing the region from −69 to −45 (PDB: 8B4E) (*12*). The ToxR molecule that binds downstream in the crystal structure (colored in pink in Fig. 6) recognizes dA −49, close to the position of TcpP #2 (colored in blue) in the cryo-EM structure. However, when the DNAs of both structures are superimposed, the ToxR and TcpP #2 recognition helices appear positioned in orthogonal orientations, since the transcription factors bind in opposite directions (Fig. 6A and fig. S11). This observation is consistent with the preferred binding site definition for each transcription factor found previously (*8*, *10*). Moreover, the binding of both transcription factors to this DNA position is exclusive, since TcpP and ToxR would clash if bound at the same time.

TcpP mutants confirm the critical residues for *toxT* promoter activation

To assess the importance of the interactions observed in our structures, we conducted a series of *toxT* promoter activation assays in *V. cholerae*, using a *toxT-lacZ* fusion reporter (*14*) and full-length TcpP mutants with a C-terminal herpes simplex virus (HSV) tag in the periplasm (Fig. 6B; Materials and Methods). The mutants Q80A, K101A, F72A, and F72S drastically reduced *toxT* activation. While Q80 is the main residue in the DNA binding helix required for *toxT* activation, E76 seems less critical (~70% of wild-type activity in E76A). Moreover, the K101A, in the wing domain, drastically reduced *toxT* activation and DNA binding (Fig. 6, B and C), in agreement with the base-specific recognition shown in the structures. The mutants contained in the TcpP transactivation loop (F72A and F72S) exhibit an impaired transcriptional activation of the reporter gene. DNA binding assays confirmed that F72A maintained *toxT* promoter binding, further indicating that the defect in this mutant is likely with RNAP α-CTD interaction (Fig. 6C). Somewhat surprisingly, TcpP derivatives with E76A, T79A, and Q80A mutations all maintained DNA binding activity similar to wild-type TcpP, suggesting that their defects in *toxT* activation are due to misaligned interaction with the promoter rather than lack of DNA binding. Furthermore, the E76A mutation appears to cause loss of specificity in DNA binding as that mutant recognizes a *toxT* promoter mutated within the TcpP binding sites (Fig. 6C).

The ToxR-TcpP (or TcpP-TcpP) interaction patch (S20, Q22, M30, P32, L35, T38 and I41) showed modest (P32S, I41S, and a

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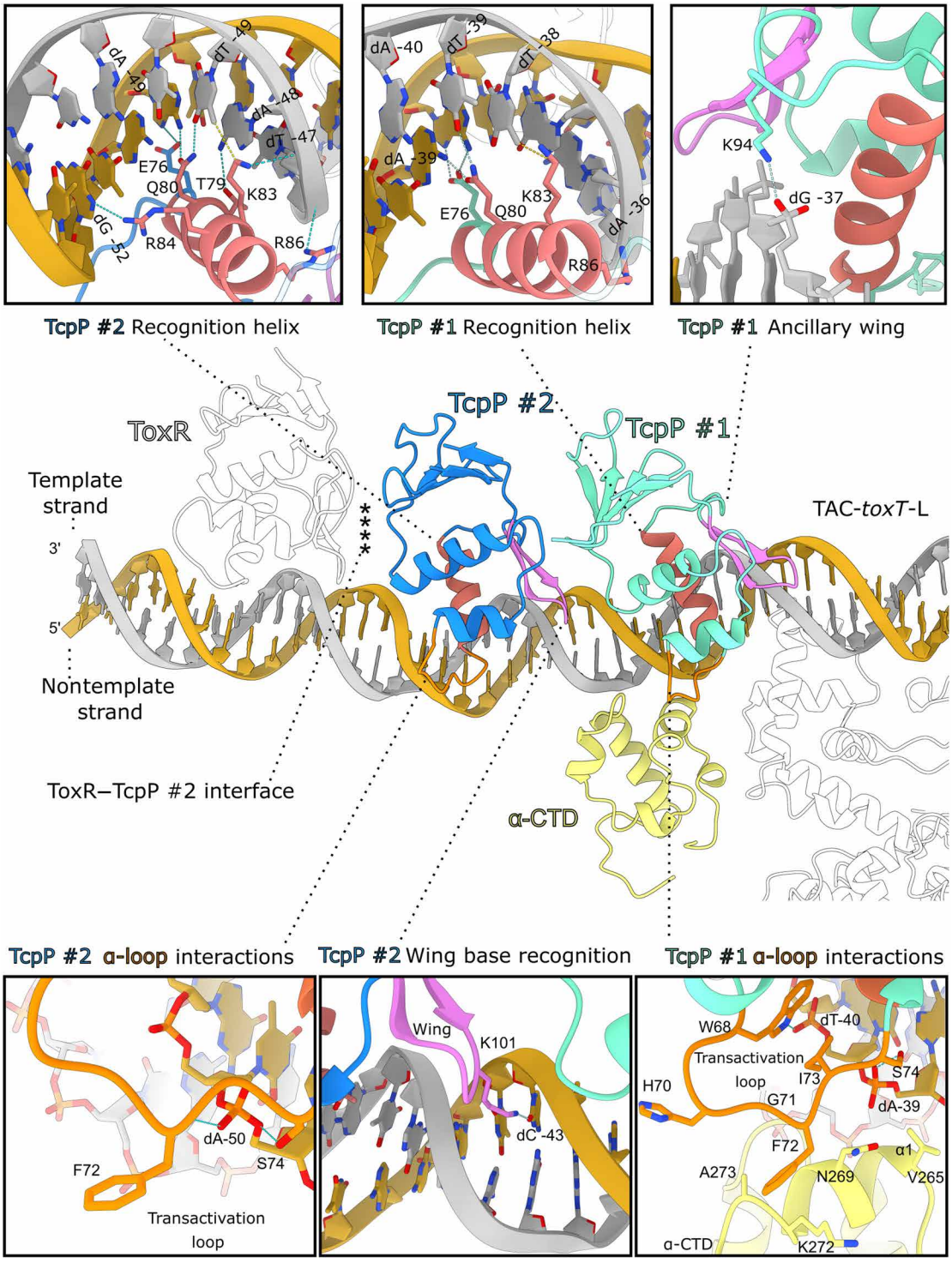

Fig. 5. TcpP also activates transcription by interacting with the α-CTD from the RNAP holoenzyme. Ribbon representation of the *toxT* transcription activation interface from the TAC-*toxT*-L complex. The subunits are colored as follows: RNAP core and ToxR, transparent; TcpP^{cyt} #1, cyan; TcpP^{cyt} #2, dodger blue; RNAP α-CTD, yellow; *toxT* template DNA strand, silver; nontemplate, gold. Recognition helices in TcpP^{cyt} protomers are colored in salmon, and the wing in violet. Top panels show the recognition helices of TcpP^{cyt} #2 and #1 (left and middle panels) with the specific base-recognition interactions and critical R86-DNA phosphate bond, plus the ancillary wing support (right panel). The bottom panels represent the TcpP^{cyt} #2 transactivation loop (or α-loop, orange) interactions with the DNA (left panel), the TcpP^{cyt} #2 wing tip (purple) base recognition (middle panel), and the transcription activation interaction (right panel) between the TcpP^{cyt} #1 transactivation loop (or α-loop, orange) and the RNAP α-CTD.

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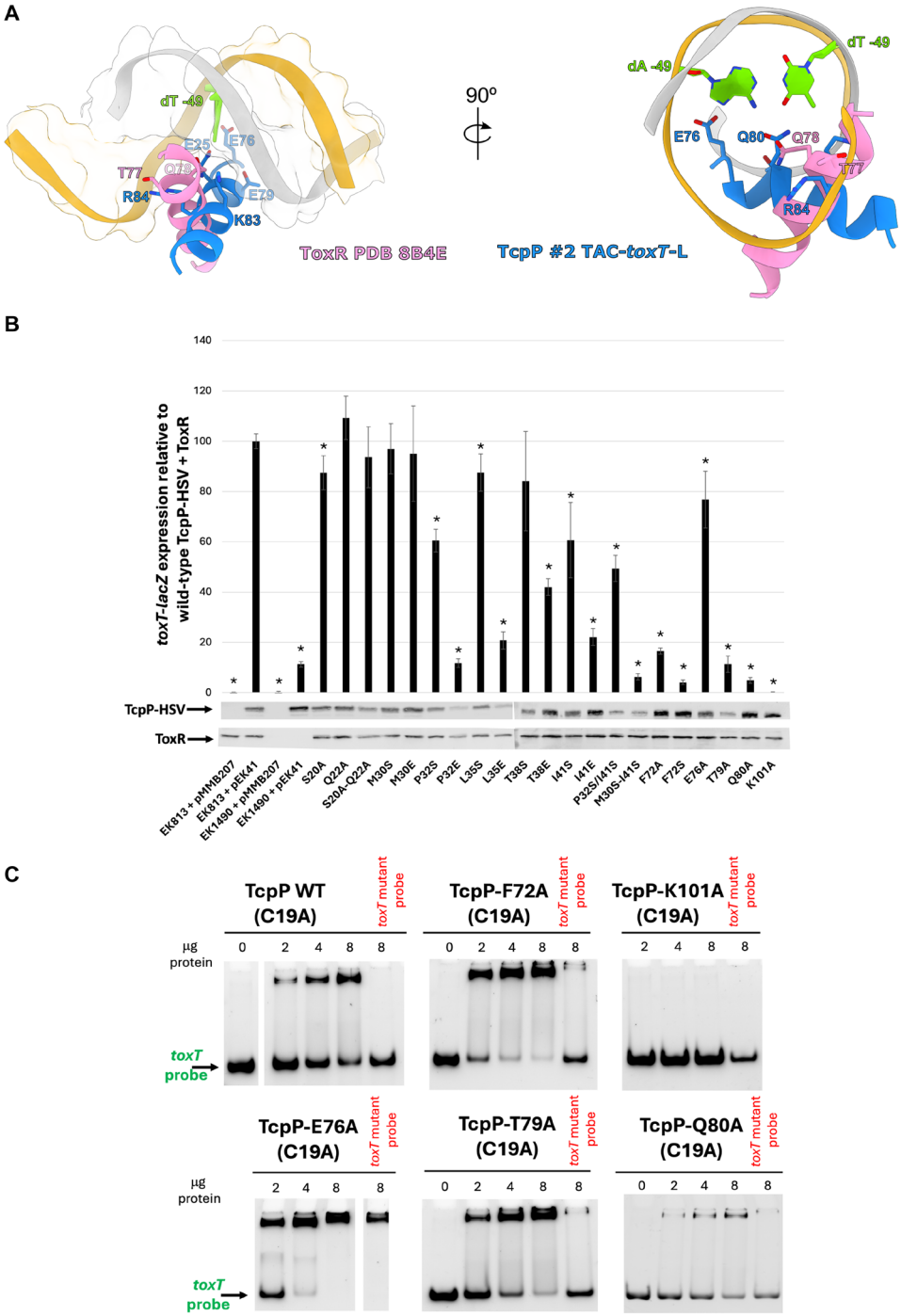

Fig. 6. The ToxR^{cyt} domain can bind the TcpP DNA operator of the *toxT* promoter in an opposite orientation. (**A**) Ribbon representation of the TAC-*toxT*-L and PDB 8B4E structure superposition of the DNA sequence. The DNA is represented in ribbon. Colors are coded as follows: nontemplate strand, gold; template strand, silver; ToxR from PDB 8B4E, pink; TcpP #2 from TAC-*toxT*-L, blue. Only the recognition helix residues recognizing the dAT −49 bp (green) are shown. (**B**) *toxT-lacZ* expression directed by TcpP mutants highlights key residues for *toxT* activation. Polar or charged residues predicted to contribute to DNA or protein interaction were changed to alanine. Residues within a hydrophobic patch from M30-I41 were changed to serine or glutamate. Changes to glutamate often rendered the TcpP protein unstable, as assessed by Western blot using a C-terminal HSV. While a quintuple serine mutant changing all contact residues from M30-I41 was completely defective for *toxT* activation (M30S-I41S), the protein was also unstable. Alternatively, a P32S/I41S double mutant was also tested. Two different substitutions of the α-CTD-contacting residue, F72, were constructed, F72A and F72S. Both were defective for *toxT* activation. T79A was also defective for *toxT* activation, but this protein was unstable. Normal levels of ToxR were present in all strains with the exception of the Δ*toxR*Δ*tcpP toxT-lacZ* strain EK1490. Strain EK813=Δ*tcpP toxT-lacZ*. $*P < 0.002$ as determined using the Student's *t* test for multiple comparisons ($P = 0.05/24$). Error bars represent the standard deviation of six biological replicates. (**C**) Purified TcpP mutants show that K101 is critical for DNA binding, while F72A has no impact on DNA binding. Single mutations in the recognition helix (E76A, T79A, and Q80A) fail to disrupt DNA binding. A *toxT* promoter with mutations in the binding site is assayed with each protein mutant to show binding specificity as a control.

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P32S/I41S double mutant) or no defects in activation. Those hydrophobic residues substituted with glutamate (E) were often unstable.

DISCUSSION

The cryo-EM data analysis of the *ompU* and *toxT* TACs revealed a common spatial disposition of their transcription factors. Both ToxR and TcpP bind in tandem at the upstream region of their respective promoters. Whereas the TAC-*ompU* presents two ToxR protomers, one of them contacting the α-CTD (Fig. 3A), the TAC-*toxT*-S contains two TcpP molecules at the same promoter position, relative to the transcription start site, and playing equivalent roles. The whole architecture is extended upstream in the *toxT* promoter with a further ToxR binding and interacting with TcpP in the TAC-*toxT*-L complex (Fig. 4).

Contrary to what was expected due to their high structural similarity with the *E. coli* PhoB transcription factor (*16*), both ToxR and TcpP DBDs interact with the RNAP α-CTD instead of with the sigma factor (Figs. 3 and 5). The α-CTD is an essential domain of the RNAP with low binding efficiency, but critical for the transcription of certain genes, showing less or nontranscriptional activity when depleted (*21*, *22*). Hence, the stabilization of this domain to the UP element of the promoter must be relevant to activate the transcription in the case of both *ompU* and *toxT* promoters. Thus, a triadic interaction involving the transcription factors, the DNA, and the α-CTD might result in the stabilization of the α-CTD binding, attaching it to the transcription complex in a more favorable state for the RNAP to start the transcription. According to our data, ToxR and TcpP would play this role during *ompU* and *toxT* activation, respectively.

The interaction of both ToxR and TcpP with the α-CTD described above unveils their functional classification as class I activators. Two transcription factor molecules bind to their respective operator, and, at the same time, the α-CTD binds to the protomer by a small interaction surface called "velcro" (*23*). In contrast to class II activators such as PhoB, neither ToxR nor TcpP produce any major visible changes in the RNAP, as stated for class I activators. The only observable change is a small adjustment in the α-CTD helix α1 and loop α3-α4 for better binding. Thus, tighter binding of the RNAP appears to be the activation mechanism rather than internal structural movements in the polymerase. For both transcription factors, a critical phenylalanine in the transactivation loop interacts with the α-CTD. The TcpP-F72 is essential for this activation (Fig. 6B), and we have previously shown that ToxR-F69A maintains DNA binding, yet it is only 35% active for *ompU* activation. Furthermore, ToxR-F69A has almost no effect on *toxT* expression, emphasizing the direct role of ToxR in *ompU* activation and an indirect role in *toxT* expression (*14*).

One of the most intriguing aspects of *toxT* transcription regulation is the different roles played by ToxR and TcpP. ToxR alone is unable to activate *toxT* transcription, but it greatly increases TcpP-mediated activation; meanwhile, TcpP can activate the transcription independently, but only when overexpressed (*11*, *24*). The structure of the TAC-*toxT*-S complex with two bound $TcpP^{cyt}$ molecules shows that $TcpP^{cyt}$ is able to bind to the *toxT* promoter in the absence of ToxR, although this occurs only when high concentrations of $TcpP^{cyt}$ are used to prepare the complex.

ToxR-DNA binding mainly involves interactions with the DNA backbone (14 out of 17 in the case of ToxR #2 at the *ompU* promoter), strengthening the idea that ToxR reads the sequence indirectly by recognizing an A-tract stretch DNA structure. In contrast, TcpP-DNA appears to be more specific, with half of the interactions (8 out of 16 in the case of TcpP #2 at the *toxT* promoter) involving direct DNA base recognition (fig. S6B). Considering the distribution of the two transcription factors along the *toxT* promoter in our TAC-*toxT*-L complex structure (Figs. 4A and 5), it appears that ToxR serves as an additional anchor point to hold tightly in place the TcpP molecules downstream and thereby facilitates both the recruitment and the proper accommodation of TcpP for interaction with the RNAP α-CTD.

A common feature shown in both *ompU* and *toxT* activation complexes is the formation of an activation complex with the α-CTD and the $\sigma^{D}{}_{4}$, with minor contacts between these two subunits of the RNAP (Figs. 3A and 5). Although the precise contribution of these transcription factor–promoted α-CTD–$\sigma^{D}{}_{4}$ interactions to the promoter activation cannot be assessed through the cryo-EM structures, it should not be underestimated in a context where fine-tuning regulation may make the difference. Thus, we cannot rule out that the interaction between the α-CTD and the σ^{D4}, stabilized by the presence of the transcription factors ToxR and TcpP, also contributes to the activation mechanism.

The ability of the $ToxR^{cyt}$ domain to bind in opposite orientations might introduce an additional *toxT* transcription regulatory mechanism. ToxR is a versatile transcription factor that activates and represses *toxT* transcription, depending on its concentration (*25*). A common feature of all our activation cryo-EM structures is the orientation of the bound transcription factors. The wing in ToxR, and in TcpP, is always positioned toward the transcription start site, facilitating the binding of other transcription factors downstream (Figs. 2A and 5). Conversely, the ToxR bound to the TcpP operator in the crystal structure presents the wing positioned backward to the transcription start site, which is not compatible with transcription (Fig. 6A and fig. S11B). If ToxR were bound in the orientation shown in the crystal, TcpP #2 would be excluded and not be able to assist the TcpP #1 binding and/or stabilization, which in turn would prevent a proper interaction with the RNAP α-CTD. The high concentration of ToxR under the crystallization conditions may have caused the binding in a nonphysiological mode. However, it is also possible that ToxR functions as an activator or as a repressor of the same promoter, depending on its binding position and orientation. As we have discussed earlier (*12*), the existence of multiple ToxR binding sites with distinct affinities in the *toxT* promoter region may improve the adaptability of *V. cholerae* to the fluctuating environmental conditions along its intestinal colonization process (*26*). It has been shown that overexpression of ToxR decreases *toxT* expression compared to endogenous levels of ToxR expression (*8*) and that ToxR expression and/or DNA binding activity responds to external stimuli (*27*). Thus, higher ToxR concentrations responding to specific environmental conditions might enable the repressing occupation of lower affinity binding sites, providing an additional fine-tuning regulation mechanism.

Our data support a common mechanism for both *ompU* and *toxT* activation in *V. cholerae*, based on the direct contacts between the RNAP α-CTD and the transcription factors (Fig. 7). There are, however, some differences that suggest a distinct gene expression regulation. The crystal structure of the ToxR-*ompU* complex (PDB: 8B4B) shows four $ToxR^{cyt}$ molecules bound to the promoter (Fig. 2D), which is incompatible with the activation state shown in the

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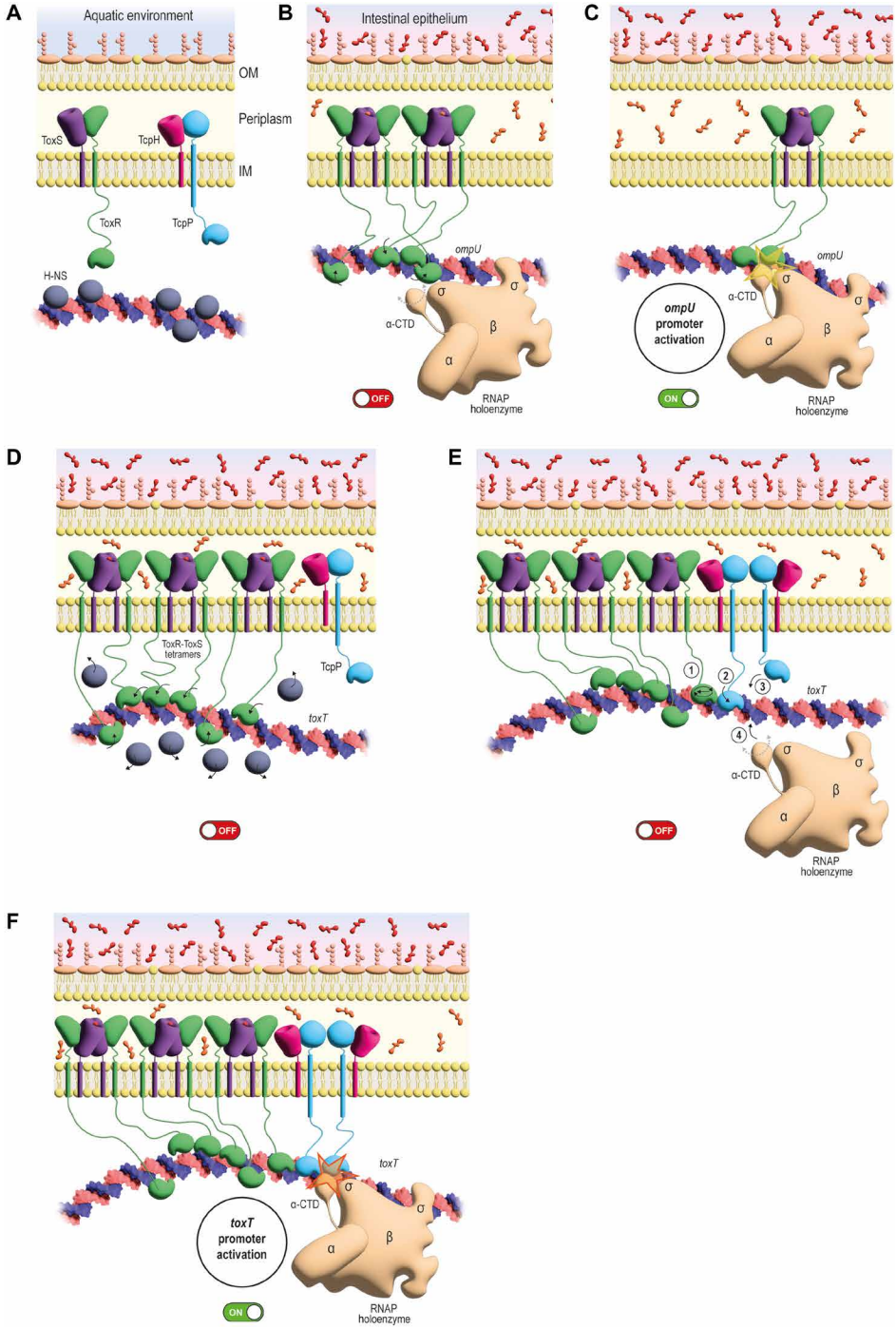

Fig. 7. Proposed mechanism for *ompU* and *toxT* activation in *V. cholerae*. (**A**) In the aquatic environment, some of the *V. cholerae* virulence genes, such as *toxT*, are repressed by H-NS proteins (*28*). (**B**) Upon arrival to the intestinal lumen, bile salts promote the formation of ToxR-ToxS complexes (*12*, *52*). The RNAP holoenzyme approaches the −35 region of the *ompU* promoter. (**C**) A stable ternary complex involving the DNA, the most downstream ToxR molecule of the tandem, and the RNAP α-CTD is formed, facilitating the *ompU* promoter activation. (**D**) At the *toxT* promoter, ToxR molecules bind immediately upstream from the transcription start site in different orientations (*12*), which removes the H-NS proteins and may recruit the *toxT* promoter to the membrane near to the TcpPH complex. As the bacterial cell progresses through the mucus layer of the intestine toward the host epithelial cells, bile concentration decreases (*26*), resulting in less functional ToxR-ToxS complexes. This positively selects the most stable, sequence-specific ToxR-DNA interactions and removes or relocates the ToxR molecules with a weaker binding, some of them probably repressing the promoter by preventing the attachment of the RNAP. Thus, a sequence of events results in the stable binding of the RNAP holoenzyme to the *toxT* promoter (**E**): first, ToxR molecules bound in unproductive or repressing positions leave their sites or are relocated ① to assist the subsequent TcpP binding; second, one TcpP molecule ② is bound around region −48; third, a second TcpP molecule ③ is bound in tandem to the contiguous downstream site and stabilized by TcpP-TcpP contacts; and fourth, the RNAP α-CTD ④ binds to this second TcpP molecule through the α loop, and, as a result, a stable DNA-TcpP-RNAP complex is formed. When this ternary complex is stable enough, the *toxT* promoter becomes activated (**F**).

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cryo-EM TAC-*ompU* complex as the RNAP α-CTD and the σ^{D}_{4} would clash with two of the ToxR^{cyt} monomers. It is unclear, with the present data, whether the binding of four ToxR^{cyt} molecules observed in the ToxR^{cyt}-DNA crystal structure is an initial stage (i.e., for better tightening of the DNA to the membrane), before the RNAP displaces two of them, or it is due to the high concentration of ToxR^{cyt} under the crystallization conditions. In contrast to what occurs in the *toxT* promoter, overexpression of ToxR does not hinder *ompU* expression, ruling out a functional repression role of the two extra ToxR^{cyt} bound.

The *toxT* activation regulation (Fig. 7C) is more complex due to the presence of both ToxR and TcpP, and the fact that histone-like nucleoid structuring (H-NS) proteins repress this promoter (*28*). According to the nature and total number of the interactions, the first TcpP molecule to bind would more probably be the most upstream, TcpP #2, (Fig. 7E). The subsequent binding event would be that of TcpP #1 (Fig. 7E), immediately downstream and showing fewer contacts with the DNA but attached through protein-protein interactions with TcpP #2. At this point, the conformation adopted by the DNA after the binding of two TcpP molecules would be the appropriate for the RNAP α-CTD (Fig. 7F) to interact with the transactivation loop of TcpP #1 to initiate the transcription. Because the genetic data show that ToxR needs to be present for any TcpP-driven activation, we propose that ToxR interaction favors the binding of TcpP #2 to the *toxT* promoter, which then facilitates the binding of TcpP #1. In addition, ToxR recruits the DNA to the membrane, facilitating the interaction of TcpP since it is localized in the membrane as well (*13*).

The cryo-EM analysis in this work has shown the underlying activation interactions and transcription factor recruitment for *V. cholerae ompU* and *toxT* transcription for host intestinal colonization. Their functional classification in this work and the detail in their respective structures allow further understanding of a previously unknown and crucial fine-tuning mechanism through the interaction between different RNAP subunits, the DNA flexibility, transcription factor orientation, DNA operator selection, and expression levels related to cell density and bile-salt concentration, among other variables. These results highlight the complexity present in *V. cholerae* transcription regulation where two transcription factors become the central point of its virulence activation. Despite the system complexity, the close match between the *E. coli* and *V. cholerae* RNAP active sites when they are superimposed opens the possibility of using current antibiotics proven effective against *E. coli*, such as rifamycins, to overcome the emergence of *V. cholerae* multidrug-resistant strains. Experiments to structurally and biochemically assess the antibiotic binding to *V. cholerae* RNAP are underway and will be presented in another paper.

MATERIALS AND METHODS

Construct design and cloning

RNAP

All genes encoding for the RNAP were polymerase chain reaction (PCR)–amplified from *V. cholerae* O395 genomic DNA, using their respective amplification oligonucleotides (*rpoA_F*, *rpoA_R*, *rpoB_F*, *rpoB_R*, *rpoC_F*, *rpoC_R*, *rpoZ_F*, and *rpoZ_R*) with the PFU Ultra II Fusion HS polymerase. The following restriction cleavage sites were added to the genes, to clone them by ligation with the T4 DNA ligase: NcoI-*rpoA*-NotI, NcoI-*rpoB*, *rpoC*-XhoI, and NdeI-*rpoZ*-PacI. Two constructs were generated for the core complex subunits and a third one for the σ^{D} factor. To avoid the formation of chimeric assemblies, the *rpoB* and *rpoC* genes, encoding for the β and β′ subunits, were fused into a single coding gene with a flexible linker (LARHVGGSGA), as described in (*29*), and a single His6 tag was added at the β′ C terminus to facilitate the purification. Therefore, the oligonucleotides *rpoB_F* and *rpoC_R* included the linker sequence, and the amplified genes were then subjected to an overlap extension PCR, with equimolar ratios of FUSION_1 and FUSION_2 oligonucleotides, to get the final *rpoBC*. The gene was digested with NcoI and XhoI enzymes and inserted into a prelinearized pET-28A vector, obtaining the plasmid pAAJ_rpoBC. Amplified *rpoA* and *rpoZ* genes were sequentially inserted into a pACYC_Duet_1 vector to obtain the final pAAJ_rpoAZ plasmid. The *rpoD* gene was PCR-amplified with *rpoD_F* and *rpoD_R* oligonucleotides and inserted into the pOPINF vector (*30*) with the In-Fusion kit, to obtain the plasmid pAAJ_rpoD.

TcpP

An *E. coli* codon optimized gene was ordered for TcpP expression of residues 1 to 114 with the Cys^{19} mutated to Ala. The gene was PCR-amplified with oligonucleotides tcpP_F and tcpP_R, including the flanking sequences required for the In-Fusion reaction. The PCR product was cloned into pOPINE vector (*30*) with a C-terminal TEV-His6 tag and obtaining the plasmid pOB_tcpP.

RNAP expression and purification

The purification protocol for *V. cholerae* RNAP core was adapted from *E. coli* RNAP purification (*31*, *32*). pAAJ_rpoAZ and pAAJ_rpoBC plasmids were cotransformed into *E. coli* BL21 (DE3). Protein expression was induced with 1 mM isopropyl-β-D-thiogalactopyranoside (IPTG) at 16°C for 72 hours. Cells were harvested and resuspended in RNi buffer [50 mM Hepes (pH 7.0), 0.5 M NaCl, and 5 mM tris(2-carboxyethyl)phosphine (TCEP)] supplemented with a cOmplete Protease Inhibitor Cocktail EDTA-free tablet and deoxyribonuclease (DNase) I (0.01 mg/ml) for lysis on a cell disruptor (Constant Systems Ltd.) at 27 kpsi (186.16 MPa) and 4°C. The lysate was centrifuged at 50,000*g* for 30 min at 4°C, and the soluble fraction was filtered and injected into a HisTrap HP 5-ml column (Cytiva). The complex was purified with a 20–column volume (CV) linear gradient from 20 to 500 mM imidazole. Fractions containing the core complex were diluted to 150 mM NaCl with Hep buffer [50 mM Hepes (pH 7.0), 5 mM dithiothreitol (DTT), and 0.5 mM EDTA] and injected into a HiTrap Heparin HP 5-ml column (Cytiva). A 5-CV isocratic step with Hep buffer at 250 mM NaCl was performed, followed by another 5-CV isocratic step at 750 mM NaCl. The peak containing the complex was diluted to 150 mM NaCl, injected into a MonoQ 5/50 (Cytiva), and washed with a 5-CV isocratic step Hep buffer at 200 mM NaCl. The complex was eluted with a mild gradient slope of 60-CV linear gradient from 200 to 600 mM NaCl. Fractions containing pure *V. cholerae* RNAP core were concentrated to a 500-μl final volume.

The σ^{D} purification protocol was adapted from *E. coli* σ^{70} (*33*). The pAAJ_rpoD plasmid was transformed into *E. coli* BL21 (DE3) pLysS strain. σ^{D} expression was induced by adding 1 mM IPTG at 30°C for 3 hours. Cells were harvested and resuspended in SNi buffer [20 mM Hepes (pH 8.0), 0.5 M NaCl, and 10% glycerol], supplemented with a cOmplete Protease Inhibitor Cocktail EDTA-free tablet and DNase I (0.01 mg/ml) for lysis on a cell disruptor (Constant Systems Ltd.) at 27 kpsi and 4°C. The lysate was centrifuged at 50,000*g* for 30 min at 4°C, and the soluble fraction was

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filtered and injected into a HisTrap HP 5-ml column (Cytiva). The protein was eluted with a 20-CV linear gradient from 20 to 500 mM imidazole. The fractions containing the protein were dialyzed in dialysis buffer [50 mM Hepes (pH 8.0), 75 mM NaCl, 0.01% Triton X-100, 10% glycerol, 0.5 mM EDTA, and 0.1 mM DTT] and injected into a MonoQ 10/100 column (Cytiva) equilibrated with Q buffer [50 mM Hepes (pH 8.0), 10% glycerol, 0.5 mM EDTA, and 0.1 mM DTT]. The protein was eluted with a 15-CV linear gradient from 75 to 500 mM NaCl Q buffer. Fractions containing the σ^D were concentrated to a final volume of 5 ml and injected into a Superdex 200 16/60 HiLoad column (Cytiva).

The *V. cholerae* RNAP holoenzyme complex was reconstructed by mixing 1:1.2 molar ratio of RNAP core and σ^D, in reconstruction buffer [10 mM Hepes (pH 8.0), 500 mM NaCl, 1 mM DTT, and 0.1 mM EDTA] for 30 min at room temperature. The mixture was injected into a Superdex 200 10/300 (Cytiva), and the eluted fractions containing the assembled complex were flash frozen and stored at −80°C.

ToxR^{cyt} and TcpP^{cyt} expression and purification

ToxR^{cyt} was expressed and purified following the protocol described in (*12*). The pOB_tcpP plasmid was transformed into BL21 (DE3) cells. TcpP expression was induced by adding 0.5 mM IPTG at 20°C for overnight expression. Cells were harvested and resuspended in PNi buffer [40 mM 2-(4-morpholino)ethanesulfonic acid (MES; pH 6.25) and 500 mM NaCl], supplemented with a cOmplete Protease Inhibitor Cocktail EDTA-free tablet and DNase I (0.01 mg/ml), for lysis on a cell disruptor (Constant Systems Ltd.) at 27 kpsi and 4°C. The lysate was centrifuged at 50,000*g* for 30 min at 4°C, and the soluble fraction was filtered and injected into a HisTrap HP 5-ml column (Cytiva). The protein was eluted using a 25-CV linear gradient from 20 to 500 mM imidazole. Fractions containing TcpP^{cyt} were concentrated to 6 mg/ml and injected into a Superdex 75 10/300 (Cytiva). The purification tags could not be removed due to protein aggregation during the cleavage process. The peak containing the pure protein was concentrated, flash frozen, and stored at −80°C.

Nucleic acid scaffold preparation

The DNA oligonucleotides were freshly annealed by mixing equimolar amounts of template and nontemplate strands, heating them at 95°C for 3 min, and immediately cooling on ice for 3 min. Afterward, the same volume of ice-cold 2× Hyb buffer [40 mM Hepes (pH 7.5), 24 mM $MgCl_2$, 200 mM NaCl, and 20 mM DTT] was added. When preparing the DNA-RNA scaffold, the RNA was preheated at 55°C for 1 min and added in 5 M excess to the preassembled double-stranded DNA (dsDNA) scaffold. The scaffold was then left at room temperature for 30 min.

Sample preparation for cryo-EM

Aliquots of *V. cholerae* RNAP at 6 mg/ml were used to prepare cryo-EM grids. All the transcription complexes were assembled by addition of 1:1.1 molar ratio of their respective nucleic acid scaffold and incubation for 10 min at room temperature. The RPo-*ompU* and TAC-*ompU* complexes were reconstructed mixing the RNAP with the T_ompU69 and NT_ompU69 oligonucleotides. The TAC-*ompU* complex further contained a 50-M excess of ToxR^{cyt}, which was incubated for 20 min. The TAC-*toxT*-S complex was prepared by incubation of the RNAP with the DNA scaffold containing the T_toxT71 and NT_toxT71 oligonucleotides, plus a 10-M excess of TcpP^{cyt}, for 20 min. The TAC-*toxT*-L was assembled mixing the RNAP with the T_toxT110 and NT_toxT110 oligonucleotides and the sequential addition of 10-M excess of ToxR^{cyt} and then TcpP^{cyt}, and incubation for 10 min at each point. The TIC-*ompU* complex was built mixing the RNAP and the T_ompU_R and NT_ompU_R oligonucleotides with the RNA_5 nucleic acid scaffold. The RPo-*ompU* and TIC-*ompU* complexes were supplemented with 4 mM 3-[(3-cholamidopropyl)dimethylammonio]-2-hydroxy-1-propanesulfonate (CHAPSO) right before vitrification (*34*). The TAC-*ompU*, TAC-*toxT*-S, and TAC-*toxT*-L complexes were diluted with complex buffer [Hepes (pH 8.0), 150 mM NaCl, and 1 mM DTT] to 0.4 mg/ml right before vitrification. Mesh 200, Cu R2/1 grids (Quantifoil) were used for the RPo-*ompU* and TIC-*ompU* complexes, whereas Mesh 200, Cu + 2 nm C R2/1 grids (Quantifoil) were used for the TAC-*ompU*, TAC-*toxT*-S, and TAC-*toxT*-L complexes. All the grids were glow-discharged using a NanoClean plasma cleaner (Fischione Instruments, Model 1070) with 75%:25% argon-oxygen mixture for 45 s. The grids were then placed into a Vitrobot Mark IV (Thermo Fisher Scientific), where 3 μl of sample was applied at 100% humidity and 8°C, blotted, and plunge frozen into liquid ethane. The blotting parameters were blot force of 0, blot time of 4 s, and incubation time of 15 s.

Data collection for the RPo-*ompU* and TAC-*ompU* complexes

The data acquisition was performed on a Titan Krios transmission electron microscope (TEM) operated at 300 keV (Thermo Fisher Scientific), equipped with a K3 direct electron detector (Gatan) and quantum energy filter using SerialEM (*35*). The magnification was 105,000× with 0.822 Å per pixel. Both datasets were acquired with a defocus range between −0.7 and −2.0 μm in counting mode. For RPo-*ompU*, 20,507 movies were recorded with 1.38 $e^-/Å^2$ per frame over 40 frames. For TAC-*ompU*, a total of 13,240 movies were recorded with 1.28 $e^-/Å^2$ per frame over 40 frames.

Data collection for the TIC-*ompU*, TAC-*toxT*-S, and TAC-*toxT*-L

The data collection for these complexes was performed on a Titan Krios G4 TEM operated at 300 keV (Thermo Fisher Scientific), equipped with a C-FEG, Selectris X energy filter and Falcon 4 direct electron detector (Thermo Fisher Scientific) using SerialEM (*35*). A magnification of 130,000× with its corresponding 0.731 Å per pixel was used. The defocus was from −0.7 to −1.8 μm, and all datasets were recorded in counting mode. For TIC-*ompU*, a total of 13,489 movies were recorded with a total accumulated dose of 50 $e^-/Å^2$. For TAC-*toxT*-S, a total of 27,113 movies were collected, with an accumulated dose of 51.26 $e^-/Å^2$. For TAC-*toxT*-L, 12,824 movies were acquired with an accumulated dose of 50.71 $e^-/Å^2$.

Cryo-EM data processing

Unless otherwise stated, the preprocessing strategy was common for all datasets. The micrographs were initially preprocessed with WARP 1.0.7W (*36*), including frame alignment, dose weighting, and contrast transfer function (CTF) estimation. The particles were picked using BoxNet2Mask 20180918 network, extracted, and subjected to a 2D classification and an ab initio reconstruction in CryoSPARC 3.0 (*37*). Thereafter, the data processing was performed in RELION 4.0. The micrographs were motion-corrected with MotionCor2 RELION (*38*) and CTF-corrected with Gctf (*39*).

The final resolution of each map was calculated using the RELION postprocessing job, based on the gold-standard Fourier shell

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correlation (FSC) with the 0.143 cutoff. The local resolution maps were generated using the local resolution tool in RELION. Last, the raw maps were sharpened using the DeepEMhancer deep-learning approach (*40*).

RPo-*ompU* data processing

The RPo-*ompU* data processing is summarized in fig. S1. Picked particle coordinates from WARP were imported to RELION and extracted in a 90-pixel-size box (binned by a factor of 4). All binned particles were submitted to a global 3D classification with a T-parameter value of 20, using the CryoSPARC Ab-Initio Reconstruction map as a reference [previously resized to a 90-pixel-size box with EMAN2 package (*41*) and low-pass filtered in RELION]. Two out of the eight 3D classes resembled an RNAP with all the subunits. Particles included in these two classes were reextracted into a 360-pixel-size box (no binning factor applied) and refined, obtaining a 3.3-Å resolution map. This map showed very low resolution at the α-CTD, which was only visible at low thresholds. We created a soft mask at the α-CTD, generated by fitting the *E. coli* RNAP structure [PDB: 7MKJ; (*17*)] with USCF ChimeraX 1.5 (*42*), and performed a masked 3D classification without alignment. After refinement of the best class particles (199,680 particles), a 3.4-Å resolution map was obtained. This set of particles was further CTF refined, polished using Bayesian polishing (*43*), CTF refined again, and subjected to a final refinement, obtaining a 3.0-Å resolution map.

TAC-*ompU* data processing

The TAC-*ompU* data processing is summarized in fig. S3. All the binned particles were extracted in RELION and equally distributed into three random subsets, which were independently subjected to a global 3D classification, with a T-parameter value of 20, using the CryoSPARC Ab-Initio Reconstruction map as a reference [previously binned five times to a 90-pixel-size box using EMAN2 (*41*) and low-pass filtered in RELION]. Particles resembling a full bacterial RNAP with an extra density in the upstream region of the DNA were joined in a single subset and 3D classified again under the same conditions. Particles from the selected class were unbinned, reextracted, and refined, leading to a 4.3-Å resolution map, which was iteratively improved by particle CTF refinement and polished using Bayesian polishing (*43*), giving a 3.1-Å map (Map A). The upstream region of the DNA showed a low-resolution density, where two $ToxR^{cyt}$ could be fitted using the PDB: 8B4B (*12*). To improve the map quality in this region, two soft masks were created using USCF ChimeraX 1.5 (*42*). The first mask covered the two N-terminal domains of the α subunit and the β, β′, and ω subunits, plus internal regions of the σ^D [using as a reference PDB: 7MKJ; (*17*)]. The second mask covered the upstream DNA, the σ^D nonconserved region, the $\sigma^D{}_4$, one α-CTD, and two $ToxR^{cyt}$. Those masks were used for the RELION multibody refinement (*44*), and two additional maps were obtained: Map B at 3.03-Å resolution and Map C at 3.77-Å resolution. A new map (Map D) was generated combining the sharpened Maps A, B and C with the "vop maximum" command in USCF ChimeraX 1.5 (*42*).

TIC-*ompU* data processing

The TIC-*ompU* data processing is summarized in fig. S2. All micrographs were motion corrected using MotionCor2 (*38*) from RELION, and CTF corrected with Gctf (*39*). crYOLO software was used for automated picking (*45*), with a 400-pixel-size box. The 1,068,380 picked particles were extracted with RELION 4.0 (*46*), exported to CryoSPARC 3.0 (*37*) and subjected to a 2D classification under the same conditions as RPo dataset. An ab initio map was reconstructed in CryoSPARC using the particles from the best 2D classes. In parallel, all picked particles were extracted in RELION with a 100-pixel-size box (binned four times) and subjected to a global 3D classification using a T-parameter value of 20 and the ab initio map from CryoSPARC as a reference [previously resized with EMAN2 (*41*) and low-pass filtered in RELION]. Particles belonging to the best class were reextracted in RELION without binning and refined. The α-CTD was only seen at low threshold values. Thus, a soft mask was applied to the α-CTD region, using the "molmap" command in USCF ChimeraX 1.5 (*42*) and the PDB: 7MKJ (*17*) as a reference model. A masked 3D classification without alignment was performed demanding only two classes. The particles belonging to the class that resembled the α-CTD were further refined, and a 3.4-Å map was obtained. Particles were CTF refined and polished with Bayesian polishing (*43*). The final refinement yielded a 2.6-Å resolution map.

TAC-*toxT*-S data processing

The TAC-*toxT*-S data processing is summarized in fig. S4. All particles were extracted in RELION and equally distributed into three random subsets, which were independently subjected to a global 3D classification, with a T-parameter value of 20, using the CryoSPARC Ab-Initio Reconstruction map as a reference [previously binned four times to a 128-pixel-size box using EMAN2 (*41*) and low-pass filtered in RELION]. Particles resembling a full bacterial RNAP with an extra density in the upstream region of the DNA were joined in a single subset and 3D classified again under the same conditions. Particles from the selected class were unbinned, reextracted, and 3D classified again into two classes. The best class included 159,379 particles and was refined to a 3.25-Å resolution map. This map was iteratively improved by particle CTF refinement and polished using Bayesian polishing (*43*), giving a 2.69-Å map (Map E). To improve the quality of the map in the upstream DNA region, two soft masks were created with USCF ChimeraX 1.5 (*42*) for the RELION multibody refinement job (*44*). The job provided two maps: Map F at 2.77-Å resolution and Map G at 3.17-Å resolution. A new map (Map H) was generated combining the sharpened Maps E, F, and G with the Phenix tool "Combine focused maps".

TAC-*toxT*-L data processing

The TAC-*toxT*-L data processing is summarized in fig. S5. All particles were extracted in RELION and subjected to a global 3D classification, with a T-parameter value of 20, using the CryoSPARC Ab-Initio Reconstruction map as a reference [previously binned four times to a 128-pixel-size box using EMAN2 (*41*) and low-pass filtered in RELION]. A total of 88,323 particles from the selected class were unbinned, reextracted, and refined to a 3.6-Å resolution map. This map was iteratively improved by particle CTF refinement and polished using Bayesian polishing (*43*) up to a 2.9-Å map (Map I). To improve the quality of the upstream region of the map, two soft masks were created with USCF ChimeraX 1.5 (*42*) and used for the RELION multibody refinement job (*44*). The job provided two maps: Map J at 2.9-Å resolution and Map K at 3.6-Å resolution. A new map (Map L) was generated combining the sharpened Maps I, J, and K with the Phenix tool "Combine focused maps". The particles for the final refinement were reextracted with a 1536 box, binned three times, and 2D classified in CryoSPARC (fig. S11). In the initial

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cryo-EM maps, the TAC-*toxT*-L complex presented an unexplained density in the upstream region of the promoter, eventually identified as the upstream DNA of an adjacent TAC-*toxT*-L complex (fig. S11A). This prompted us to repeat the particle extraction using a larger-sized box. The subsequent map, refined at low resolution (7.3 Å), confirmed that two TAC-*toxT*-L TACs form a C2 symmetry dimer on our grids (fig. S11A). The density corresponding to the upstream DNA region has three protomers for each RNAP monomer. The two protomers closest to the polymerase core are clearly $TcpP^{cyt}$ molecules, determined both at the TAC-*toxT*-S and TAC-*toxT*-L maps. The density for this third protomer in the low-resolution map of the TAC-*toxT*-L dimer allowed a rigid-body fit of a $ToxR^{cyt}$ molecule, and the resulting orientation was further translated into the TAC-*toxT*-L monomeric structure. Particles clearly belonging to dimer conformation were subjected to nonuniform refinement, symmetry C2 expansion, and, last, local refinement, yielding an FSC 7.2-Å resolution map.

Model building

The TIC sharpened map was selected to start the model building of the RNAP subunits, due to its higher resolution. The AlphaFold predictions (*47*) of all *V. cholerae* O395 RNAP subunits were downloaded, and rigid body fitted into the TIC sharpened map, using USCF ChimeraX 1.5 (*42*). The *E. coli* RNAP PDB: 7MKJ (*17*) was used as a guide to build the DNA bubble, which was mutated to match the promoter sequence. The resulting model was then rigid body fitted in Coot 0.9 (*48*). Moreover, an ideal B-DNA with the *ompU* promoter sequence was generated in Coot 0.9 (*48*) and real space refined into the upstream and downstream density map outside the transcription bubble, applying strong restraints. A single-stranded ideal RNA was generated in Coot 0.9 (*48*) and rigid body fitted into the density. All the complex structure was then rigid body refined by subunits and real space refined with strong restraints. Afterward, the model was manually rebuilt and adjusted, residue by residue in Coot (*48*), using both the sharpened and unsharpened maps. The structure was iteratively refined against the sharpened map with Phenix 1.13 Real-Space Refinement for Cryo-EM (*49*), with the secondary structure restraints option turned on.

The TAC-*ompU* was built using the final TIC PDB as a template but removing the RNA chain and the upstream DNA region. An ideal B-form DNA, matching the sequence of the designed oligonucleotides, was generated in Coot 0.9 (*48*) and fitted as explained above. All the RNAP subunits were rigid body fitted to the sharpened Map A in Coot (*48*). Two $ToxR^{cyt}$ monomers from the PDB: 8B4B structure (*12*) were rigid body fitted into the corresponding densities. All subunits were individually real space refined to the density, applying strong restraints. The model was rebuilt and adjusted manually in Coot (*48*), using the sharpened and unsharpened Maps A, B, C, and D. Last, the model was iteratively refined in Phenix 1.13 (*49*) Real-Space Refine tool against Map A, and the last two rounds against Map D (due to the poorly resolved regions in Map A), with secondary structure restraints enabled.

The RPo-*ompU* structure was built by removing the $ToxR^{cyt}$ subunits from TAC-*ompU* structure and rigid body fitting the new model into the RPo-*ompU* map. Each subunit was real space refined into the density with strong restraints in Coot 0.9 (*48*). Afterward, the model was rebuilt and adjusted manually against the sharpened and unsharpened maps, and, lastly, iteratively refined in Phenix 1.13 (*49*).

The TAC-*toxT*-S building followed the same steps as the TAC-*ompU*, where the DNA was mutated to match the new promoter sequence and the $ToxR^{cyt}$ protomers were replaced by the TcpP full-length model downloaded from AlphaFold, where the Cys^{19} residue was mutated to Ala, and only the cytoplasmic domain $TcpP^{cyt}$ (residues 3 to 114) was rigid body fitted into the density. The structure was manually adjusted at the residues where clear density was visible. The refinement was performed in Phenix, applying strong secondary structure restraints, against Map E and against Map H, for the last refinement iteration, due to the poor-resolved regions.

To build the TAC-*toxT*-L structure, the TAC-*toxT*-S was used as a template, where the $ToxR^{cyt}$ presence was detected as a poorly resolved extra density. The structure was built and iteratively refined against Map I in Phenix, using the Real-Space Refine tool. Subsequently, the structure was fitted into the dimer map density, and a $ToxR^{cyt}$ molecule was added to its corresponding density. Last, the structure comprising the two $TcpP^{cyt}$ and the $ToxR^{cyt}$ was rigid body fitted against the Map I in Phenix.

Bacterial strains used

V. cholerae classical strain O395 was the original strain for all strains used in this study. EK813 is an O395 derivative carrying a chromosomal *toxT-lacZ* fusion and a deletion of chromosomal *tcpP* allele (Δ*tcpP*) (*10*). Strain EK1490 is O395 Δ*toxR*Δ*tcpP toxT-lacZ* and lacks both ToxR and TcpP, the two regulators required for *toxT* expression (*10*).

Construction of TcpP mutants

Oligonucleotides encoding mutations of interest in the *tcpP* gene were designed and ordered from Thermo Fisher Scientific (table S3). Oligonucleotides were incorporated into a plasmid encoding a C-terminally HSV-tagged allele of *tcpP* by PCR [pMMB207-tcpP-HSV (pEK41); (*10*)]. Following PCR, the template plasmid was digested with DpnI for 2 hours at 37°C, and the resulting PCR product was transformed into *E. coli* DH5α. Candidate chloramphenicol-resistant colonies were grown, plasmid DNA was isolated, and plasmids were sent for sequencing at Genewiz. Resulting sequences were aligned by MegAlign Pro in the Lasergene Suite of programs and clones with only the desired nucleotide changes throughout the entire *tcpP* coding region saved. Each *tcpP* mutant encoded on pEK41was then introduced into the *V. cholerae* Δ*tcpP toxT-lacZ* reporter strain EK813 by triparental mating between DH5α, EK813, and an *E. coli* strain carrying the conjugatable plasmid pRK2013 (*50*). Transconjugants streptomycin-resistant (indicating *V. cholerae*) and chloramphenicol-resistant (carrying pEK41) were selected and tested for the ability of that TcpP derivative to activate *toxT-lacZ* expression.

β-galactosidase assays

To assess *toxT-lacZ* expression activated by the various TcpP mutants, β-galactosidase assays were performed as described previously (*10*). Briefly, *V. cholerae* strains carrying various *tcpP*-HSV mutant alleles were grown overnight in 3 ml of LB medium at 37°C, and then, overnight cultures were diluted 1:50 into 3 ml of Vc LB (LB with 50% NaCl) with 1 mM IPTG that had been adjusted to pH 6.5 at 30°C to ensure maximal virulence gene expression. Cultures were grown for about 4 hours until the optical density at 600 nm reached ~0.5, and then, β-galactosidase assays were performed on 20 μl of cultures as previously described.

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toxT promoter binding assays

To assess the DNA binding of various TcpP mutants, we overexpressed the cytoplasmic forms of mutants as C-terminally His6-tagged derivatives from vector pET30b in the *E. coli* Rosetta strain with plasmid pLysS after induction overnight at 20°C with 1 mM IPTG. All mutants also included a C19A mutation to prevent protein aggregation, as was used in the cryo-EM structure. Proteins were purified by nickel-affinity chromatography, and gel shift assays were performed on fluorescein amidite–labeling dsDNA probes representing the native TcpP binding site (TcpP binding site TOP and TcpP binding site mutant in table S3) as described previously (*12*).

Supplementary Materials

This PDF file includes:

Figs. S1 to S11

Tables S1 to S3

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Structures of *Vibrio cholerae* transcription complexes reveal how ToxR and TcpP recruit the RNA polymerase and activate virulence genes

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